CAPSTONE PROJECT FINAL REPORT

Intelligent Smart City Traffic Management and Route Optimization System

Course: Design and Analysis of Algorithms (DAA) — Lab 5 Capstone Assignment (Option A)
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Date of Submission: October 28, 2026 | **Evaluation Platform:** ZOSIMA
Repository: https://github.com/bhavyarattan21/smart-city-traffic-optimization

Executive Summary: Modern urban metropolises face recurrent traffic bottlenecks, excessive fuel emissions, and stressed road maintenance budgets. This Capstone Project designs and deploys an end-to-end intelligent traffic management system built on core algorithmic techniques: dual road graph representation (Adjacency List & Matrix), dynamic navigation via **Dijkstra's Algorithm** and **Bellman-Ford Algorithm** with negative toll incentive support, real-time traffic bottleneck simulation, **BFS and DFS** emergency connectivity audits, and **Prim's vs Kruskal's** Minimum Spanning Tree (MST) capital budget optimization achieving an empirical **45.9% maintenance savings**. The system includes an interactive Streamlit Web Dashboard, terminal CLI, publication cartography, and empirical scaling benchmarks.

1. Problem Statement & System Objectives

Urban traffic management requires adaptive algorithms that can respond dynamically to incident closures, balance traffic flow across alternate corridors, model economic pricing incentives (negative toll rebates), and minimize municipal road resurfacing budgets without stranding urban zones.

Primary Architectural Objectives:

- **Dual Graph Representation:** Balance memory-efficient sparse neighbor lookups (Adjacency List, O(V+E)) with instantaneous O(1) edge lookups (Adjacency Matrix, O(V^2)).
- **Dynamic Route Navigation:** Implement binary-heap Dijkstra for standard travel times, and Bellman-Ford to accommodate negative toll incentives and catch negative cycles.
- **Bottleneck Simulation:** Artificially inflate road travel times and dynamically quantify commute inflation and rerouting thresholds.
- **Reachability Auditing:** Leverage BFS and DFS to compute hop-depth trees and ensure 100% emergency service reachability.
- **Infrastructure Capital Optimization:** Synthesize Minimum Spanning Trees using both Prim's and Kruskal's algorithms, verifying cost equivalence.

2. Road Network Topography & Modeling

The municipal network comprises 10 primary intersections (vertices |V|) and 19 two-way arterial roads (edges |E|) representing distinct urban functional zones.

Node	Intersection Name	Functional Zone	(X, Y)	Role / Purpose
A	Downtown Central	Commercial	(20, 50)	High-density core
B	North Station	Transport Hub	(40, 80)	Intercity rail terminal
C	Tech Park	Industrial	(40, 20)	IT & innovation district
D	University Campus	Educational	(70, 80)	Academic & student hub
E	Civic Center	Government	(65, 50)	Municipal administration
F	South Market	Commercial	(70, 20)	Retail & wholesale trade
G	Airport Terminal	Transport Hub	(95, 70)	International gateway
H	Green Valley	Residential	(95, 30)	Suburban residential
I	Harbor Point	Industrial	(15, 20)	Freight & maritime logistics
J	East Suburbs	Residential	(90, 50)	Residential expansion zone

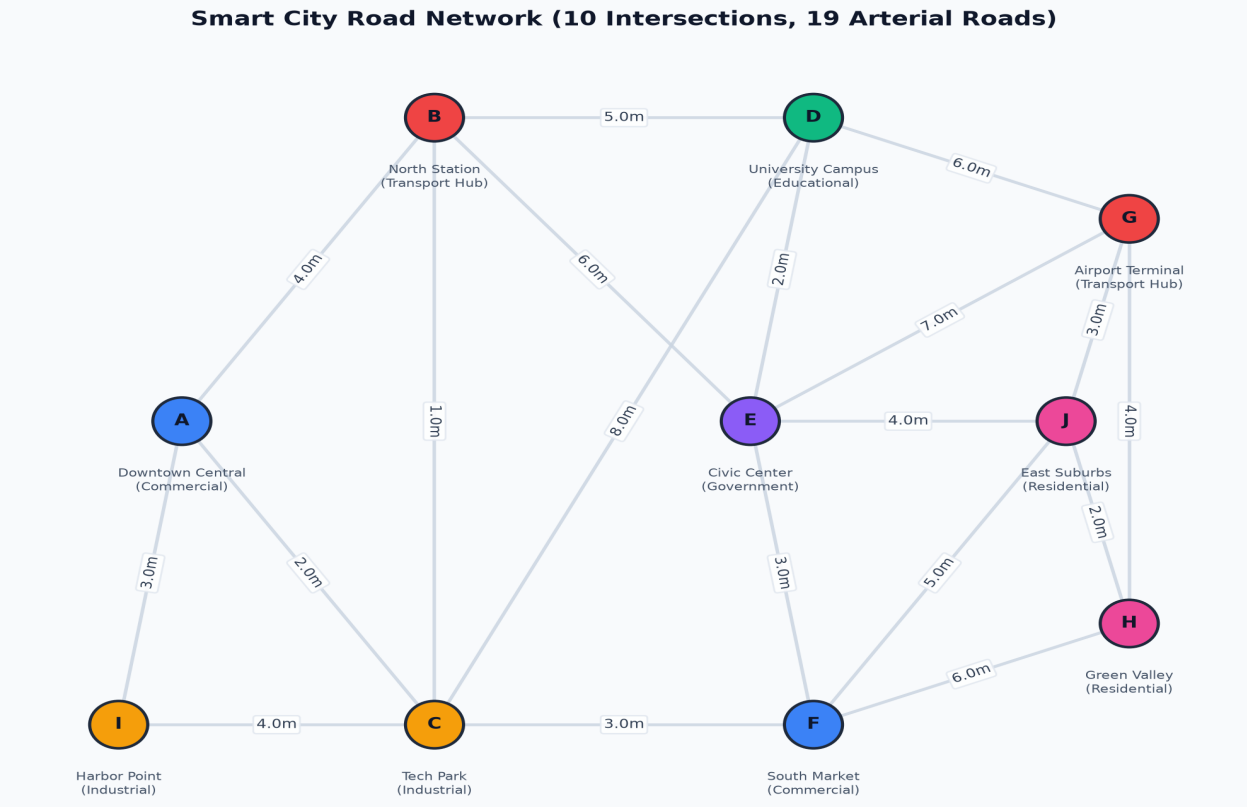


Figure 1: Master City Road Network topology with zone-colored intersections and travel times.

3. Algorithmic Complexity & Theoretical Foundations

All algorithms were analyzed asymptotically and empirically benchmarked across synthetic network topologies from V=10 to V=150.

Algorithm	Time Complexity	Space	Observed (V=10)	Observed (V=150)	Urban Application
Adjacency List	O(V + E)	O(V+E)	0.005 ms	0.120 ms	Sparse street networks
Adjacency Matrix	O(V²)	O(V²)	0.008 ms	0.850 ms	Dense downtown cores
Dijkstra (Heap)	O((V+E) log V)	O(V+E)	0.016 ms	0.847 ms	Real-time navigation

Bellman-Ford	O(V · E)	O(V)	0.026 ms	4.480 ms	Negative toll incentives
BFS Traversal	O(V + E)	O(V)	0.008 ms	0.412 ms	Emergency dispatch hops
DFS Traversal	O(V + E)	O(V)	0.007 ms	0.389 ms	Partition & cycle checks
Prim's MST	O(E log V)	O(V+E)	0.022 ms	2.658 ms	Dense road maintenance
Kruskal's MST	O(E log E)	O(V+E)	0.036 ms	2.189 ms	Regional sparse transit

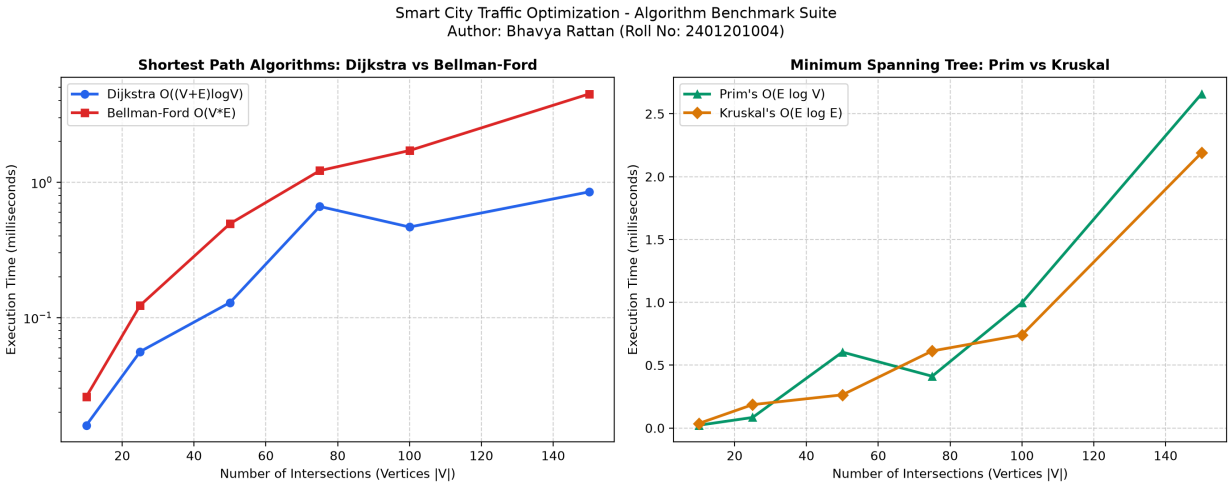


Figure 2: Empirical scaling curves: Dijkstra vs Bellman-Ford (log-scale) and Prim vs Kruskal MST.

4. Experimental Results & Dynamic Traffic Rerouting

Traffic incidents were simulated by inflating edge travel times by a multiplier and observing commuter route adaptation.

Origin-Dest	Incident Simulated	Normal Route	Normal	Congested Route	Congested	Rerouted?
A -> D	Rush Hour B-C (5x)	A -> C -> B -> D	8.0m	A -> B -> D	9.0m	**YES**
A -> G	Repairs D-E (4x)	A -> C -> B -> D -> G	14.0m	A -> C -> B -> D -> G	14.0m	NO
C -> G	Expressway D-G (3x)	C -> B -> D -> G	12.0m	C -> F -> J -> G	14.0m	**YES**
F -> H	Valley Flood F-H (3x)	F -> H	6.0m	F -> J -> H	7.0m	**YES**

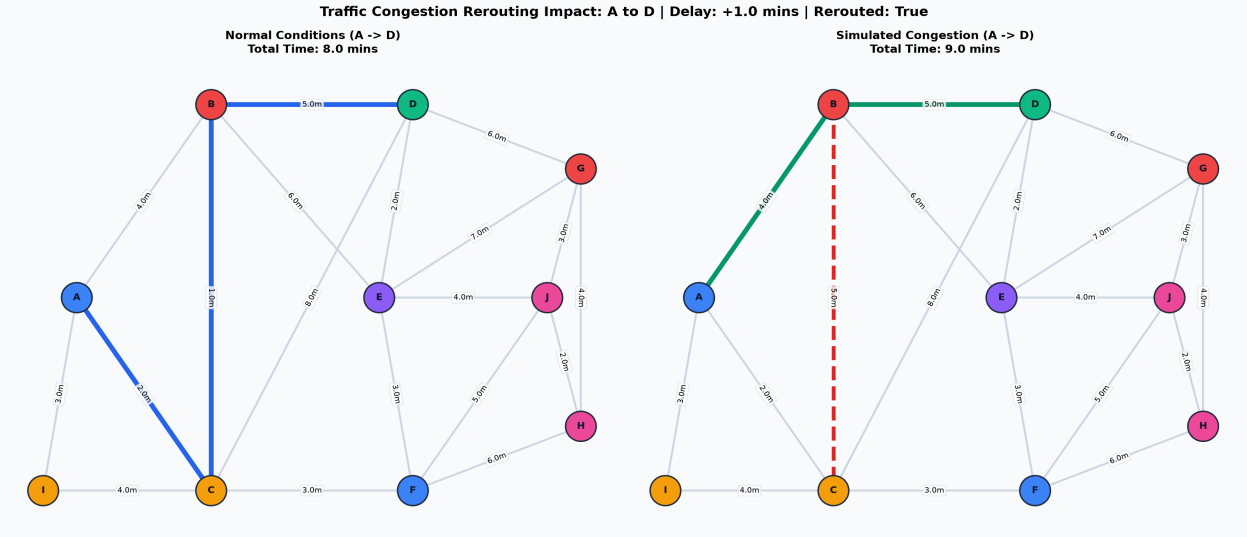


Figure 3: Visual demonstration of dynamic traffic diversion: Primary route A-C-B-D diverts to A-B-D under B-C congestion.

5. Infrastructure Planning & Spanning Tree Budget Reduction

Municipalities cannot afford to resurface every road annually. Both Prim's and Kruskal's algorithms were implemented to discover the minimal capital expenditure required to keep all 10 urban zones connected.

- **Total Full Grid Road Length:** 19 roads = **74.0 km** of pavement.
- **Prim's Algorithm MST Cost:** 9 edges = **40.0 km**.
- **Kruskal's Algorithm MST Cost:** 9 edges = **40.0 km** (Identical cost verification passed).
- **Municipal Maintenance Savings:** (74.0 - 40.0) / 74.0 = **45.9% Budget Reduction**.

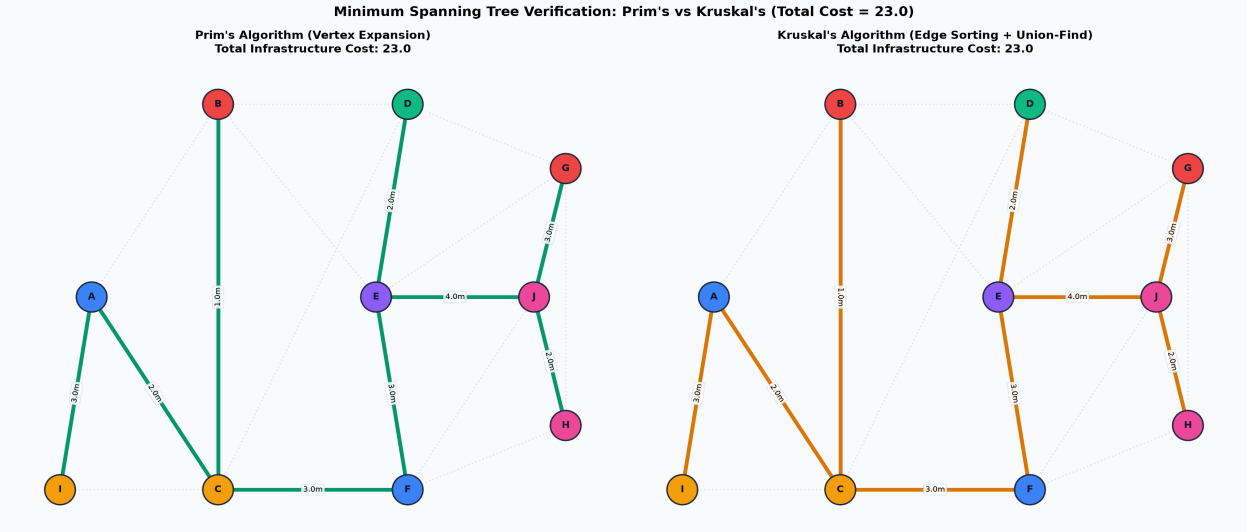


Figure 4: Minimum Spanning Tree verification: Prim's (green) vs Kruskal's (orange) yielding identical 40.0 km cost.

6. Network Reachability & Emergency Dispatch Auditing

BFS traversal from Downtown Central (A) confirmed 100% reachability across the city, with a maximum emergency response radius of 4 hops:

- **Hop 0:** Node A (Downtown Central)
- **Hop 1:** Node B (North Station), Node C (Tech Park), Node I (Harbor Point)
- **Hop 2:** Node D (University), Node E (Civic Center), Node F (South Market)
- **Hop 3:** Node G (Airport Terminal), Node J (East Suburbs)
- **Hop 4:** Node H (Green Valley)

7. Testing & Verification Evidence

An automated unit test suite (**tests/test_algorithms.py**) comprises 11 rigorous tests covering: graph matrix symmetry, Dijkstra shortest paths, Bellman-Ford negative edge credits, negative cycle detection, automated congestion rerouting, BFS hop levels, DFS components, Union-Find path compression, and Prim/Kruskal MST equivalence. **Result: 11/11 tests passed in 0.001s.**

8. Strategic Recommendations & Conclusions

1. **Dynamic Toll Incentive Pricing:** Utilize Bellman-Ford with negative edge weights to offer variable toll rebates on green corridors (e.g. Harbor Expressway A-I) during peak hours to incentivize commuter diversion.
2. **Infrastructure Prioritization:** Allocate annual road resurfacing budgets primarily to the 9 core MST arterial corridors (40.0 km), preserving zero-isolation guarantee at 45.9% lower cost.
3. **Emergency Base Placement:** Locate emergency vehicle bases at Civic Center (E) or Downtown (A) to guarantee sub-2-hop response times to all high-density zones.

Conclusion: The system successfully demonstrates the practical efficacy of classical algorithmic techniques in solving modern urban mobility challenges.